

```
1. df <- data.frame(
  name = c("Anastasia", "Dima", "Michael", "Matthew", "Laura", "Kevin",
"Jonas"),
  score = c(12.5, 9.0, 16.5, 12.0, 9.0, 8.0, 19.0),
  attempts = c(1, 3, 2, 3, 2, 1, 2)
)
print(df)
View(df)
```

Output:

	name	score	attempts
1	Anastasia	12.5	1
2	Dima	9.0	3
3	Michael	16.5	2
4	Matthew	12.0	3
5	Laura	9.0	2
6	Kevin	8.0	1
7	Jonas	19.0	2

```
2. df$qualify <- c("yes", "no", "yes", "no", "no", "no", "yes")
print(df)
View(df)
```

Output:

	name	score	attempts	qualify
1	Anastasia	12.5	1	yes
2	Dima	9.0	3	no
3	Michael	16.5	2	yes
4	Matthew	12.0	3	no
5	Laura	9.0	2	no
6	Kevin	8.0	1	no
7	Jonas	19.0	2	yes

```

3. new_row <- data.frame(
      name = "Emily",
      score = 14.5,
      attempts = 1,
      qualify = "yes"
    )
df <- rbind(df, new_row)
print(df)
View(df)

```

Output:

	name	score	attempts	qualify
1	Anastasia	12.5	1	yes
2	Dima	9.0	3	no
3	Michael	16.5	2	yes
4	Matthew	12.0	3	no
5	Laura	9.0	2	no
6	Kevin	8.0	1	no
7	Jonas	19.0	2	yes
8	Emily	14.5	1	yes

```

4. str(df)
df$qualify <- as.factor(df$qualify)
summary(df)

```

Output:

```

> str(df)
'data.frame': 8 obs. of 4 variables:
 $ name      : chr  "Anastasia" "Dima" "Michael" "Matthew" ...
 $ score     : num  12.5 9 16.5 12 9 8 19 14.5
 $ attempts  : num   1 3 2 3 2 1 2 1
 $ qualify   : chr   "yes" "no" "yes" "no" ...
>
> df$qualify <- as.factor(df$qualify)
> summary(df)
      name      score      attempts      qualify
Length:8      Min.   : 8.00    Min.   :1.000    no :4
Class :character 1st Qu.: 9.00    1st Qu.:1.000    yes:4
Mode  :character Median :12.25    Median :2.000
              Mean  :12.56    Mean   :1.875
              3rd Qu.:15.00    3rd Qu.:2.250
              Max.   :19.00    Max.   :3.000

```

```
cat("Number of rows:", nrow(df), "\n")
```

```
cat("Number of columns:", ncol(df), "\n")
```

Output:

```
> cat("Number of rows:", nrow(df), "\n")
```

```
Number of rows: 8
```

```
> cat("Number of columns:", ncol(df), "\n")
```

```
Number of columns: 4
```